DATASET DETAILS

AI-Powered Cybersecurity Incident Response Assistant

CIC-IDS2017: source, format, key fields, purpose and the preprocessing performed for retrieval and machine learning

Project	Capstone Project 8 · CIC-IDS2017
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1. Dataset summary

Property	Value
Dataset name	CIC-IDS2017 – Intrusion Detection Evaluation Dataset
Publisher	Canadian Institute for Cybersecurity (CIC), University of New Brunswick
Primary link	https://www.unb.ca/cic/datasets/ids-2017.html
Alternative source	Kaggle mirror of CIC-IDS2017 (clean Parquet version), as referenced in the project brief
Reference	I. Sharafaldin, A. H. Lashkari, A. A. Ghorbani, "Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization", ICISSP 2018
Release used	`MachineLearningCVE` : 8 CSV files of labelled network flows (CICFlowMeter features)
Format	CSV (one row per bidirectional network flow); the dataset is also published as PCAP
Size	2,830,743 flows · 79 columns · 15 traffic labels (BENIGN + 14 attack types)
Capture	Five working days (Monday–Friday, July 2017) of a realistic network with benign user traffic and scheduled attacks

2. Source files

File	Day and content	Flows
`Monday-WorkingHours.pcap_ISCX.csv`	Monday: benign traffic only	529,918
`Tuesday-WorkingHours.pcap_ISCX.csv`	Tuesday: FTP-Patator, SSH-Patator	445,909
`Wednesday-workingHours.pcap_ISCX.csv`	Wednesday: DoS (slowloris, Slowhttptest, Hulk, GoldenEye), Heartbleed	692,703
`Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv`	Thursday morning: web brute force, XSS, SQL injection	170,366
`Thursday-WorkingHours-Afternoon-Infilteration.pcap_ISCX.csv`	Thursday afternoon: infiltration	288,602
`Friday-WorkingHours-Morning.pcap_ISCX.csv`	Friday morning: botnet (Bot)	191,033
`Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv`	Friday afternoon: port scan	286,467
`Friday-WorkingHours-Afternoon-DDos.pcap_ISCX.csv`	Friday afternoon: DDoS	225,745
Total		**2,830,743**

The files are placed in `MachineLearningCVE/` (not in git). A small committed sample is in `data/sample_data/` (see §7).

3. Key fields

3.1 Fields named in the brief and their availability

The brief lists the key fields of the full CIC-IDS2017 release. The `MachineLearningCVE` release used here contains the flow statistics and labels but **not** the identity fields:

Field	In this release	Used for
Destination Port	✅	Metadata filter, incident text, prioritisation context (not a model feature)
Flow Duration, Total Fwd / Backward Packets, Total Length of Fwd / Bwd Packets	✅	Model features, incident text, numeric search filters
Flow Bytes/s, Flow Packets/s, Flow IAT Mean	✅	Flow Packets/s is a model feature and a search filter; the others are in the raw table
Packet Length Mean, Average Packet Size	✅	Model features (Packet Length Mean)
SYN / RST / ACK Flag Count	✅	Model features
Active Mean, Idle Mean	✅	Model features
Label	✅	Ground truth for classification, evaluation and search filters
Flow ID, Source IP, Source Port, Destination IP, Protocol, Timestamp	❌ not in this release	Filters implemented, but have no data (see §6.3)

3.2 The 24 features used by the models and the assistant

Feature (internal name)	CIC-IDS2017 column	Meaning
`flow_duration`	Flow Duration	Total flow duration, microseconds
`total_fwd_packets`	Total Fwd Packets	Packets from source to destination
`total_bwd_packets`	Total Bwd Packets	Packets from destination back to source
`total_len_fwd`	Total Length of Fwd Packets	Bytes sent forward
`total_len_bwd`	Total Length of Bwd Packets	Bytes sent backward
`flow_packets_s`	Flow Packets/s	Packets per second over the flow
`fwd_packets_s`	Fwd Packets/s	Forward packets per second
`bwd_packets_s`	Bwd Packets/s	Backward packets per second
`fwd_packet_length_mean`	Fwd Packet Length Mean	Mean forward packet size, bytes
`bwd_packet_length_mean`	Bwd Packet Length Mean	Mean backward packet size, bytes
`min_packet_length`	Min Packet Length	Smallest packet, bytes
`max_packet_length`	Max Packet Length	Largest packet, bytes
`packet_length_mean`	Packet Length Mean	Mean packet size (both directions), bytes

Feature (internal name)	CIC-IDS2017 column	Meaning
`packet_length_std`	Packet Length Std	Standard deviation of packet size, bytes
`syn_flag_count`	SYN Flag Count	1 if a TCP SYN flag was seen
`ack_flag_count`	ACK Flag Count	1 if a TCP ACK flag was seen
`rst_flag_count`	RST Flag Count	1 if a TCP RST flag was seen
`psh_flag_count`	PSH Flag Count	1 if a TCP PSH flag was seen
`fin_flag_count`	FIN Flag Count	1 if a TCP FIN flag was seen
`down_up_ratio`	Down/Up Ratio	Backward ÷ forward packets
`init_win_bytes_forward`	Init_Win_bytes_forward	Initial TCP window, forward, bytes (−1 = not set)
`init_win_bytes_backward`	Init_Win_bytes_backward	Initial TCP window, backward, bytes (−1 = not set)
`active_mean`	Active Mean	Mean active time before idle, microseconds
`idle_mean`	Idle Mean	Mean idle time before active, microseconds

3.3 Labels

Label	Category	Flows after deduplication	In the incident library
BENIGN	Normal traffic	1,820,651	52,210
DoS Hulk	Denial of service	166,335	7,819
DDoS	Distributed denial of service	127,188	7,843
PortScan	Reconnaissance	90,743	7,828
DoS GoldenEye	Denial of service	10,281	5,246
DoS Slowhttptest	Denial of service	5,172	2,584
DoS slowloris	Denial of service	4,867	2,634
FTP-Patator	Brute force (FTP)	4,259	4,259
SSH-Patator	Brute force (SSH)	3,149	3,149
Bot	Botnet	1,936	1,936
Web Attack - Brute Force	Web attack	1,426	1,323
Web Attack - XSS	Web attack	651	651
Infiltration	Infiltration	36	36
Web Attack - Sql Injection	Web attack	20	20
Heartbleed	Exploit	11	11
Total		**2,236,725**	**97,549**

4. Purpose in this project

Use	How the dataset supports it
Historical incident similarity (RAG)	Each sampled flow becomes an incident with an ID and a text description; analysts search them with hybrid retrieval
Hybrid search and metadata filtering	Attack label and destination port as metadata; flow statistics as numeric filters ("more than 10000 packets/s")
Attack-type classification	15 labelled classes for a supervised classifier
Intrusion-risk prediction	BENIGN vs attack labels for a calibrated risk model
Anomaly detection	BENIGN flows only, to learn what normal traffic looks like
Explainability	Flow statistics with clear meanings, explained with SHAP and compared with normal-traffic medians
Prioritisation	Model outputs plus attack-type severity → P1–P4
Evaluation	Historical labels are the ground truth for the evaluation cases and the exact checks

5. Preprocessing performed

5.1 Incident library (PySpark, `scripts/data/prepare_incident_data.py`)

Step	What happens	Rows
Load	8 CSVs read with PySpark and aligned to one schema (79 raw columns → internal names)	2,830,743
Normalise columns	Headers with stray spaces and variants (`Destination Port` / `Dst Port`) mapped to snake_case names	
Cast	40 numeric columns cast to double	
Infinity → null	`±inf` values (common in Flow Bytes/s and Flow Packets/s) set to null	
Standardise labels	Labels trimmed; the broken Web Attack separator (`Web Attack` `Brute Force`) repaired; 15 canonical names	
Deduplicate	SHA-256 hash over 11 flow-identity fields; duplicate flows dropped	2,236,725
Drop incomplete	Rows without a label or destination port removed (none in this release)	2,236,725
Stratified sample	Per-label targets (BENIGN 52,000, large attack classes about 7,800, smaller ones kept whole), seed 42	97,549
IDs and text	Shuffled, given IDs `INC-000001` … `INC-097549` , and a template-based `incident_text` (no LLM; missing values omitted)	97,549
Validate	Required columns, unique IDs, non-empty texts, no duplicates, label distribution	
Output	`data/processed/incidents_100k.parquet` (41 columns)	
Index	BM25 keyword index and all-MiniLM-L6-v2 embeddings in ChromaDB, aligned and verified against the Parquet	

Example incident text (one flow = one document, 241 tokens at most, so it is never split):

> *Attack: DoS slowloris. Destination port: 80. Flow duration: 3001067 microseconds. Forward packets: 3. Backward packets: 0. Forward bytes: 0. Backward bytes: 0. Flow packets per second: 0.999644. …*

5.2 Training table for the models (`ml/eda/data_cleaning.ipynb`)

Starting from the incident library:

1. Dropped the all-null identity columns (IPs, source port, protocol, timestamp) and constant columns.
2. Dropped redundant per-direction flag columns.
3. Dropped rows with a negative flow duration (1 row).
4. Filled the few missing `flow_packets_s` values with the median.
5. Removed `incident_id` and `incident_text` .

Result: `data/cleaned_network_data/cleaned_network_data.csv` , with 97,548 rows: label, `dst_port` and the 24 features.

Further steps in the training notebooks:

- `dst_port` **is excluded** from the model features, so the models learn traffic behaviour rather than "port X means attack Y".
- **Duplicate feature vectors are removed before** the train/test split, so no flow appears in both.
- Splits are stratified, with seed 42 (the risk model also keeps a separate split for probability calibration).
- The anomaly detector uses an arcsinh transform and is trained on BENIGN flows only.

6. Data quality notes and limitations

6.1 Class imbalance

BENIGN is 81% of the deduplicated flows, and Heartbleed (11), SQL injection (20) and Infiltration (36) are extremely rare. Sampling keeps every rare flow and caps the large classes, and the classifier uses balanced class weights. Macro F1 (0.839) is reported next to accuracy (0.986) because the rare classes are hardest.

6.2 Duplicates

594,018 duplicate flows (21%) were removed. Without this step, identical flows could appear in both the training and test sets and inflate accuracy.

6.3 Missing identity fields

This release has no IP, protocol, source port or timestamp values. The metadata filters for these fields are implemented and validated against the fields actually populated. Asking for them returns a clear message instead of an empty result. The `GeneratedLabelledFlows` release of CIC-IDS2017 includes them and could be used to enable these filters.

6.4 Single-flow incidents

Each incident is one flow. Behaviour across many flows (many sources to one target, slow scans over hours) is not visible in a single record. The assistant states this in its limitations.

6.5 Dataset artifacts

The risk model relies partly on `init_win_bytes_forward`, which is known to separate classes partly because of how the dataset was generated. SHAP makes this visible; it is documented as a limitation.

7. Sample data in the repository

`data/sample_data/` (committed):

File	Contents
`sample_incidents.csv`	120 incidents, the first 8 per label: ID, label, port, 24 features, incident text
`triage_queue_sample.csv`	14 rows (incident IDs, metric rows, a partial flow) for the Triage Queue and `POST /ml/prioritize/csv`
`sample_queries.json`	8 assistant questions with expected facts and real answer excerpts

Regenerate them with `python scripts/data/make_sample_data.py`.